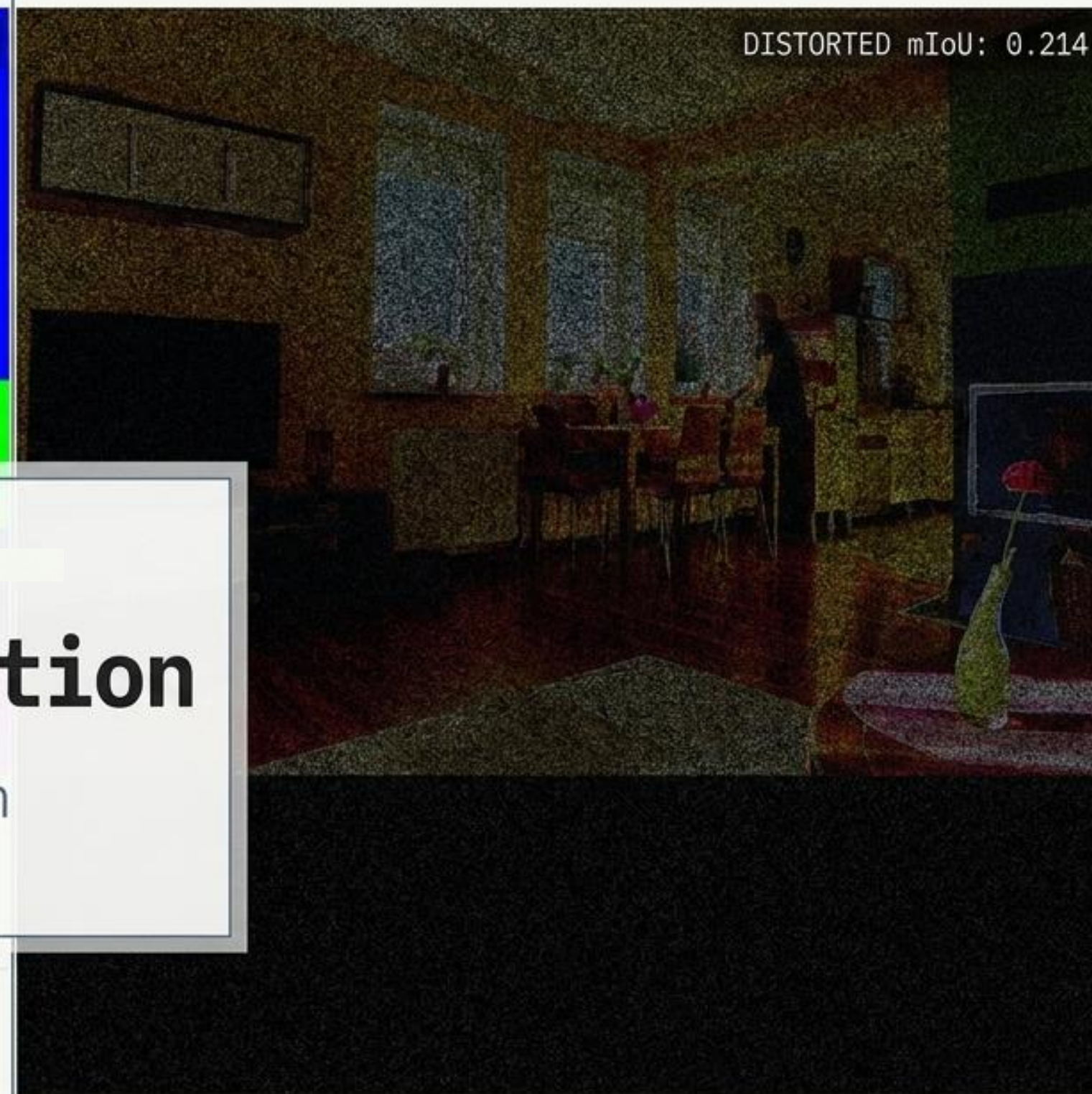
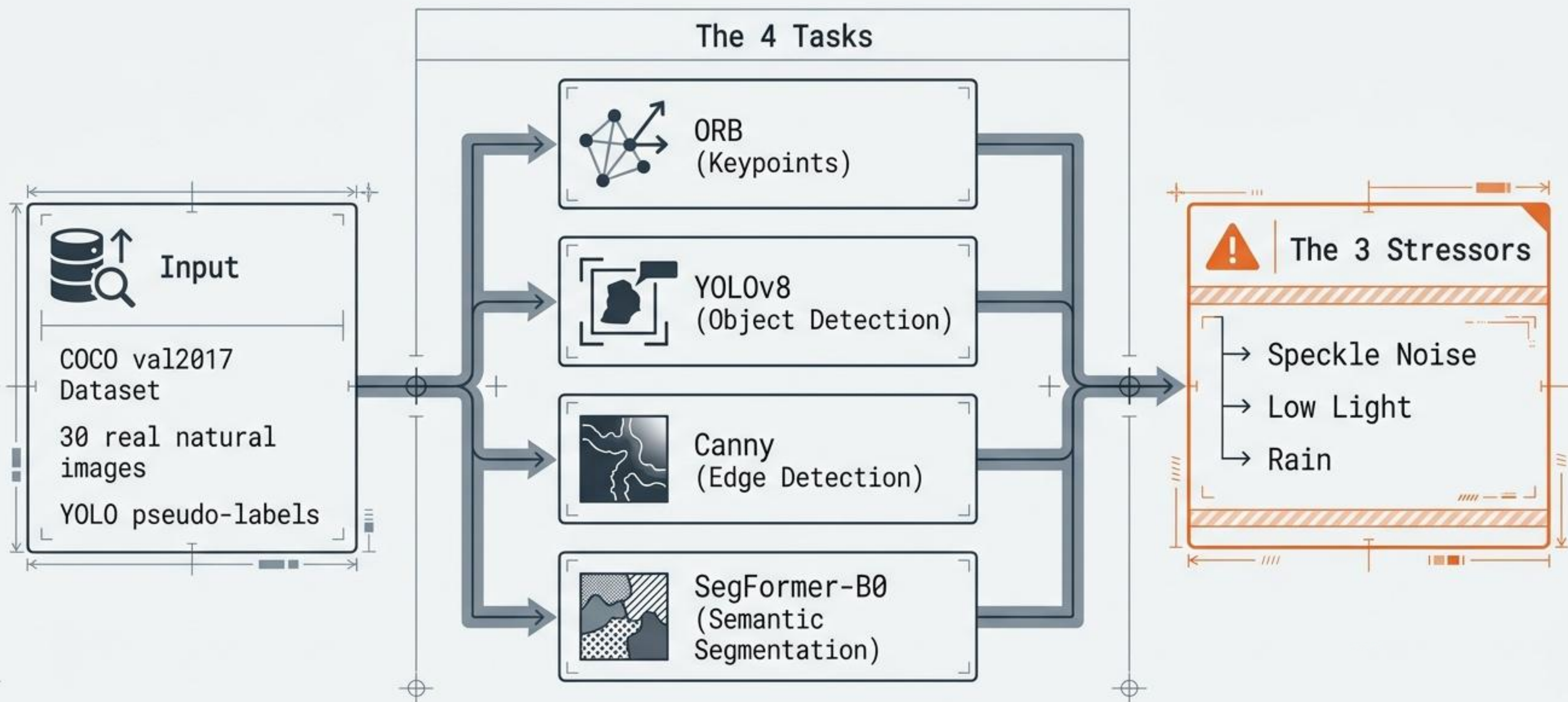




Image Processing — Robustness Evaluation

Stress-Testing Computer Vision
Algorithms Under Distortion





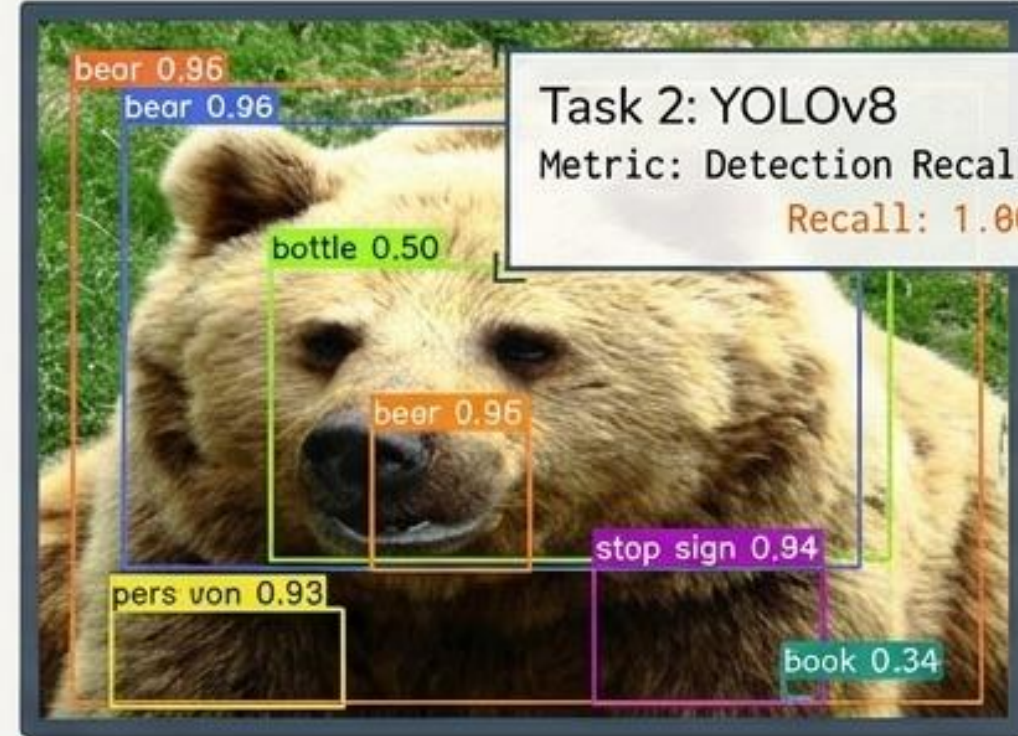
Goal: Measure algorithmic degradation under environmental stress and test recovery strategies.

BASELINE ALGORITHM PERFORMANCE (PERFECT CONDITIONS)



Task 1: ORB

Metric: Mean Keypoint Count
Value: ~800



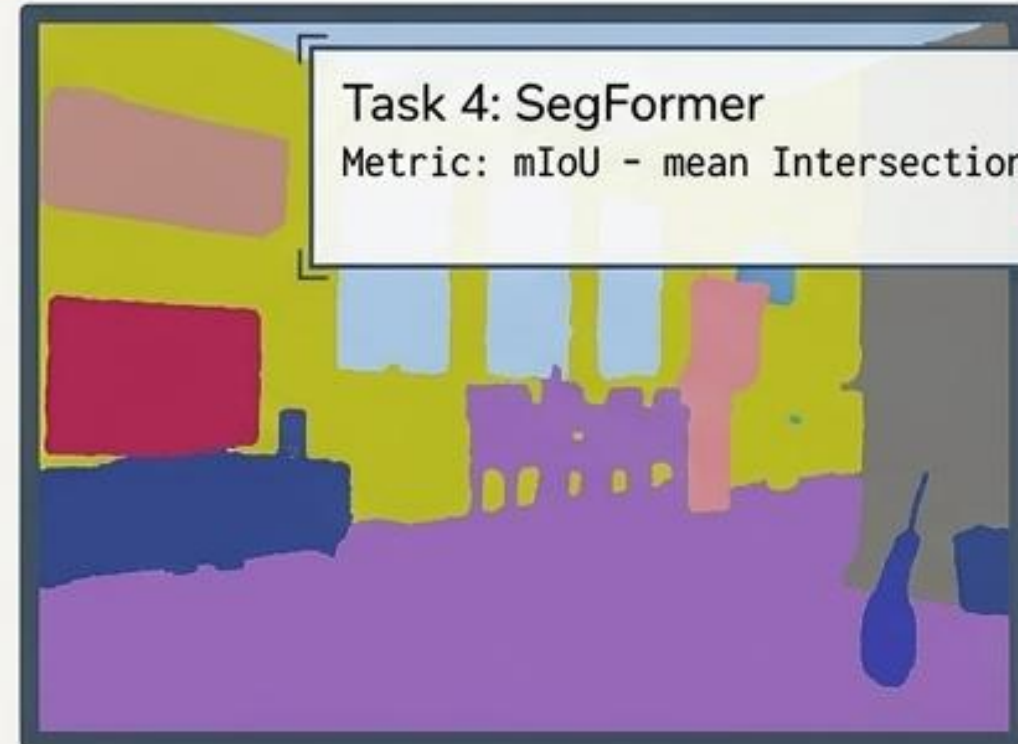
Task 2: YOLOv8

Metric: Detection Recall, $\text{IoU} \geq 0.5$
Recall: 1.00, IoU: >0.8



Task 3: Canny

Metric: Edge Density Ratio
Value: 12.5%



Task 4: SegFormer

Metric: mIoU - mean Intersection over Union
mIoU: 1.000

Goal: Establish peak performance benchmarks across all tasks using pristine input data.

The Stressors: Introducing Environmental Distortion



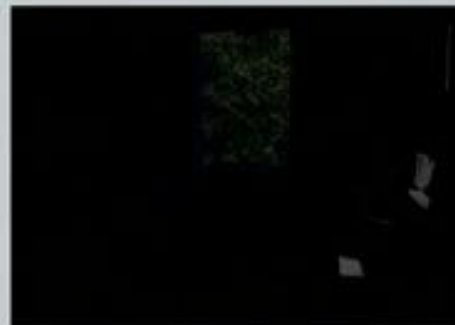
Speckle Noise

Multiplicative | Severity: Heavy



Low Light

Brightness Reduction | Severity: Severe Dark



Rain

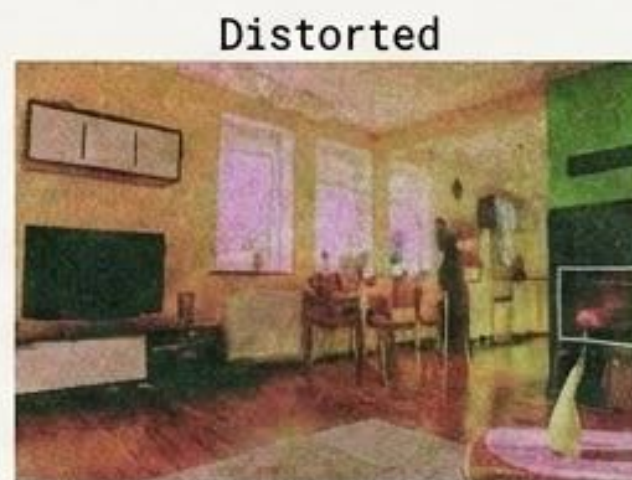
Visual Streaks | Severity: Moderate-Heavy



The Recovery Pipeline: Enhancement Strategies

Denoising

Bilateral Filter
+ Morphological
Opening



Brightening

Gamma Correction
($\gamma=0.35$) + CLAHE



De-Raining

Median Blur +
Bilateral Filter



Goal: Use targeted algorithms to restore distorted data for reliable analysis.

Degradation Drop: The Initial Impact

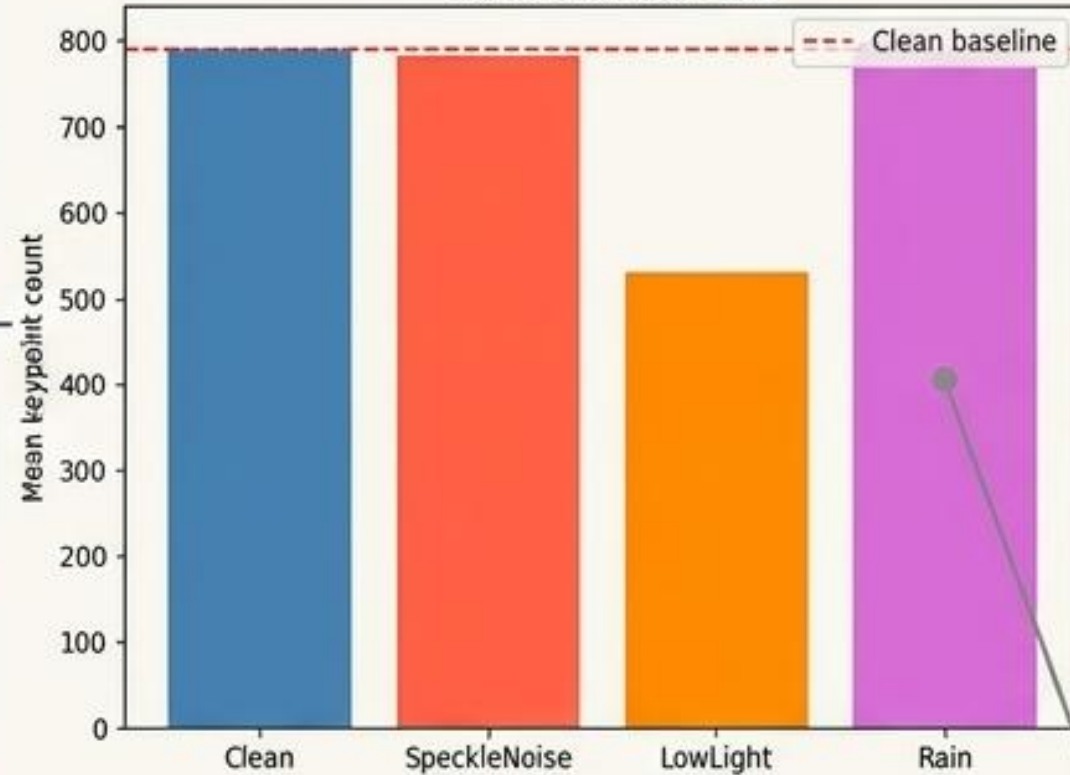
⚠ CRITICAL FAILURE

LowLight causes massive collapse in YOLO recall (-62.9%) and Edge density (-77.4%).

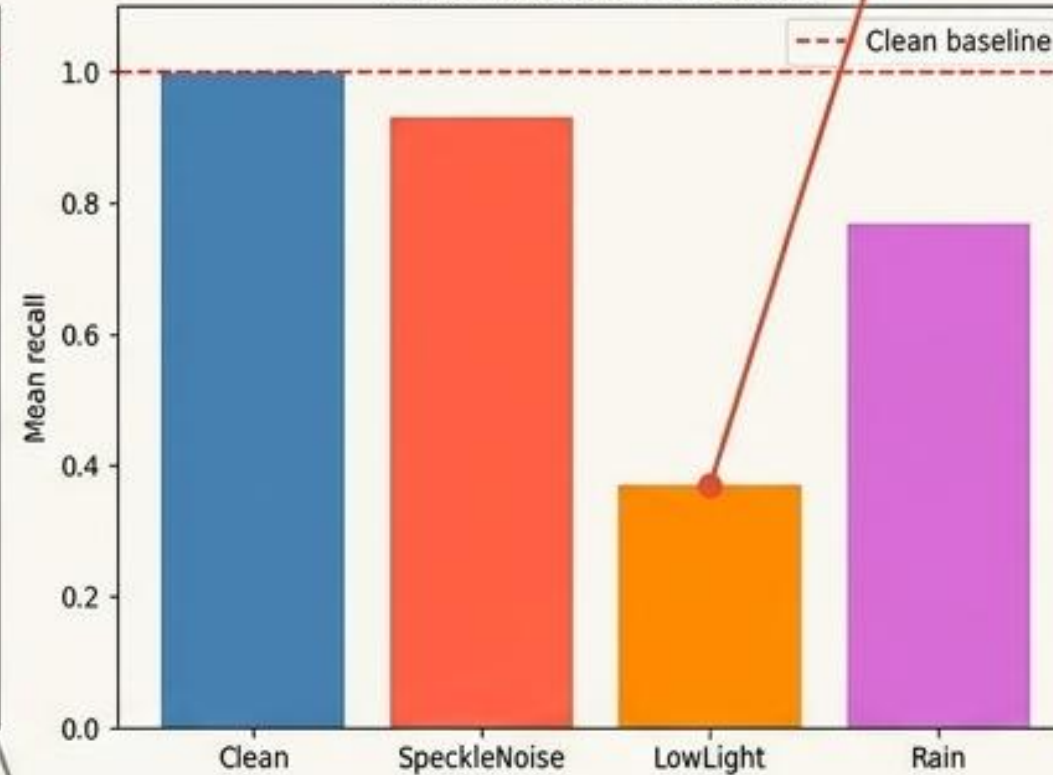
⚠ CRITICAL FAILURE

LowLight causes massive collapse in YOLO recall (-62.9%) and Edge density (-77.4%).

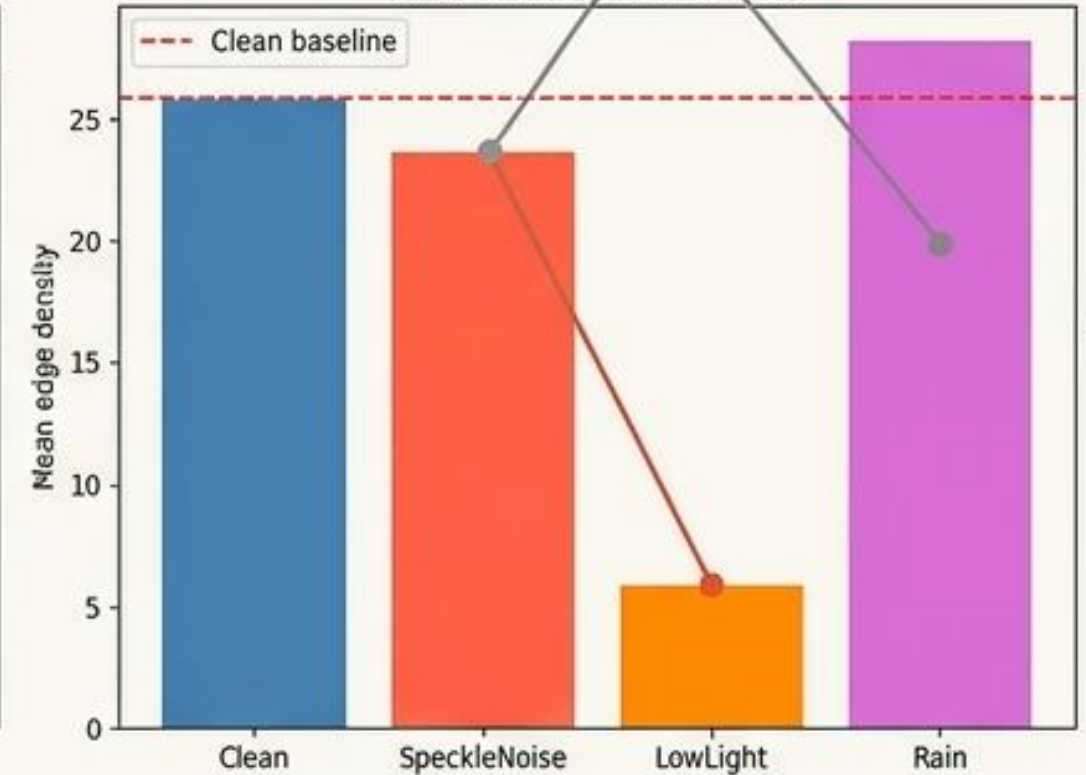
Task 1: ORB Keypoints



Task 2: YOLO Detection Recall



Task 3: Canny Edge Density



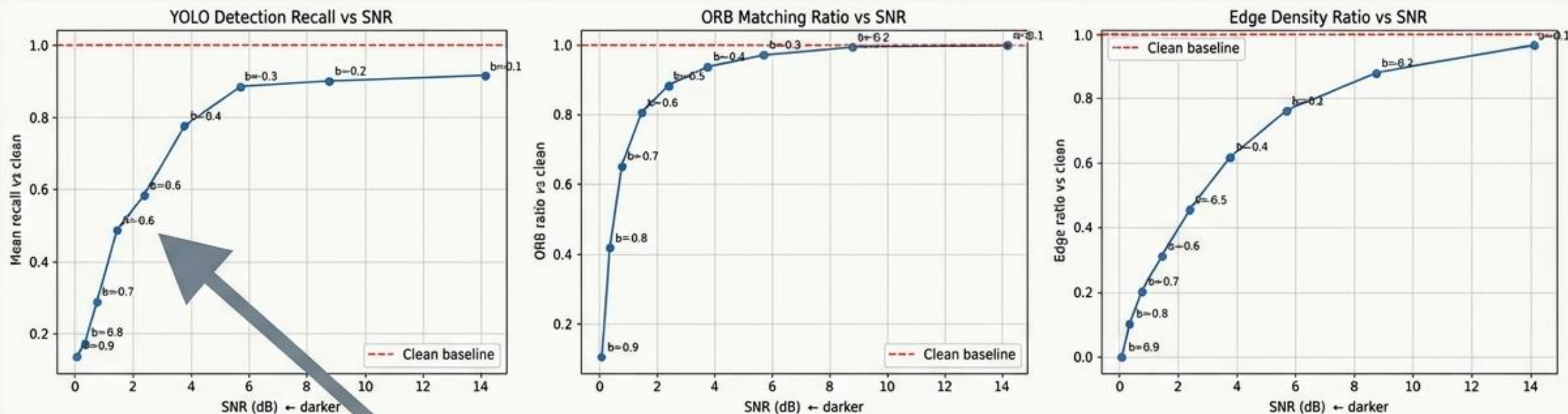
🔗 ANOMALY DETECTION

Anomaly: Rain streaks artificially inflate keypoint and edge counts.

✅ ROBUSTNESS WINNER

Semantic segmentation (SegFormer) remains stable, dropping <9% across all stressors.

The Darkest Hurdle: Brightness vs. Performance



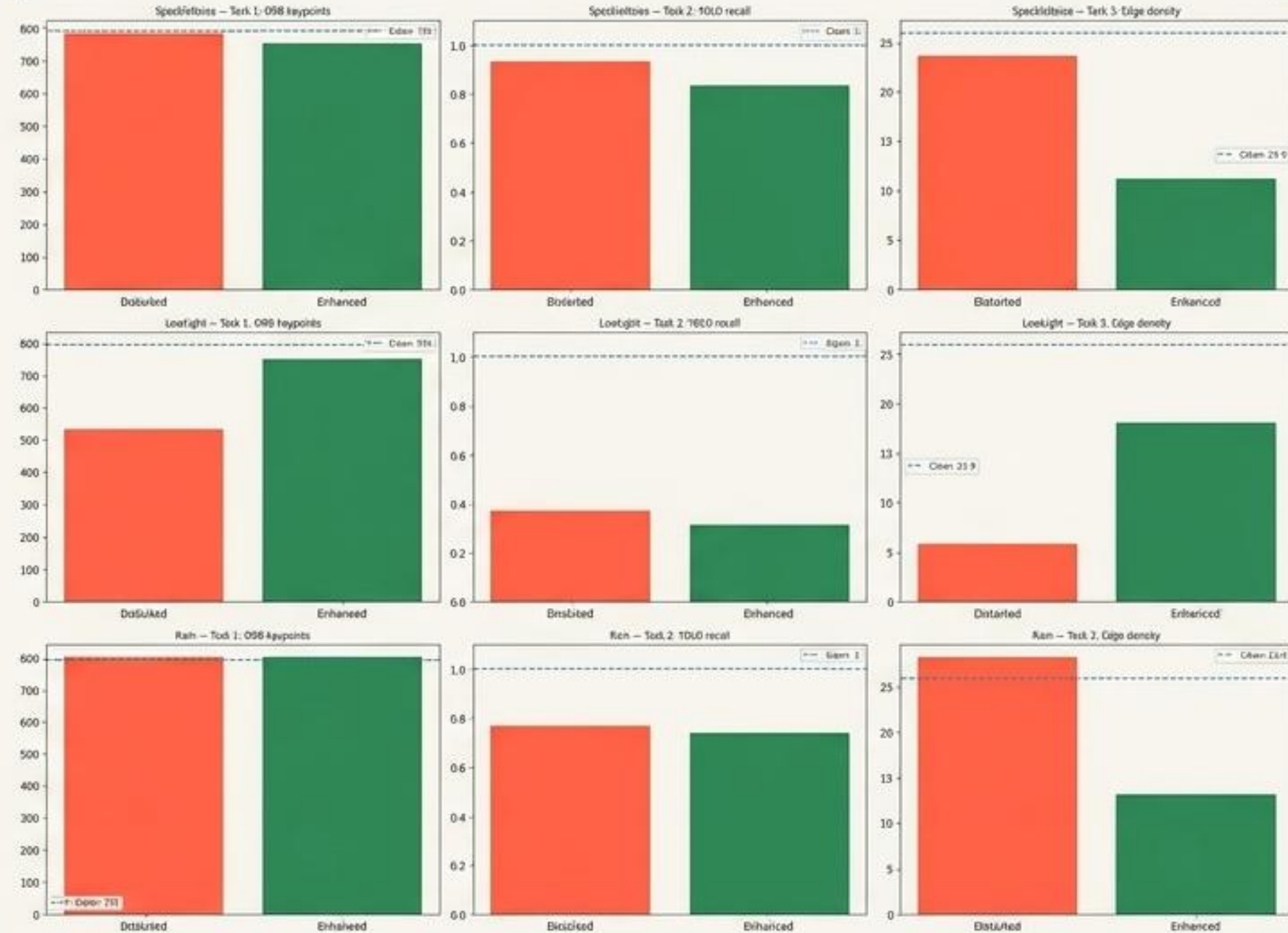
Clear monotonic degradation.

As SNR drops toward zero, YOLO recall and Edge density vanish entirely. Brightness reduction is the single most destructive distortion.

Enhancements: A Double-Edged Sword

✓ THE SUCCESS

Gamma + CLAHE effectively recovers keypoints (+41%) and edge density (+208%) under LowLight conditions.

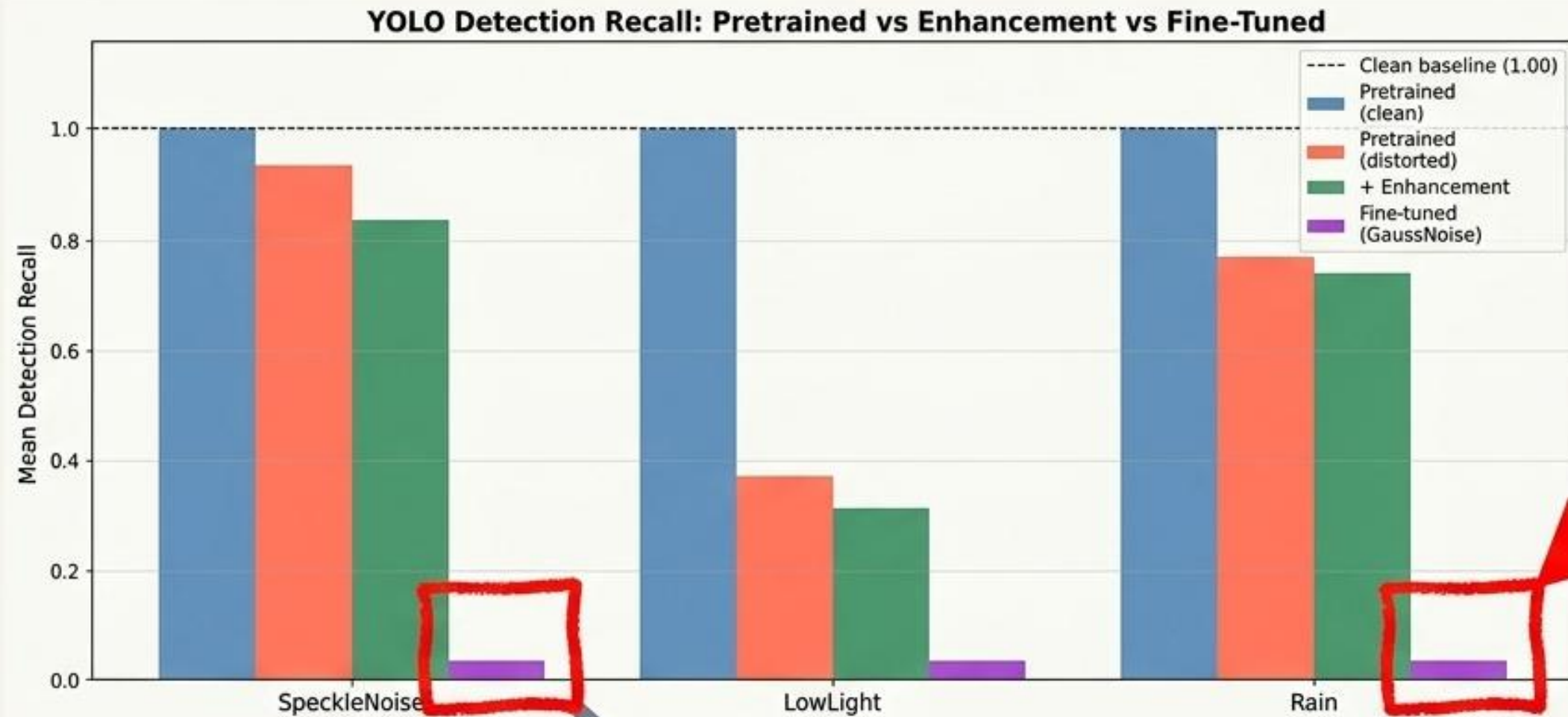


! THE COST

Filtering removes spurious noise but smooths out real object details, causing YOLO recall to drop even further (-10.4% for noise).

NOTE: Segmentation (mIoU) remains unaffected by either distortion or enhancement.

The Catastrophic Forgetting Anomaly



Recall collapsed to 0.033 across ALL distortions.

Diagnostic Insight:

Fine-tuning YOLO on a limited dataset (5 epochs / 29 images) destroyed its pretrained representations.

Naive, low-data fine-tuning is actively worse than doing nothing.

Final Verdict: The Robustness Matrix



Most Damaging Distortion: Low Light

Devastates pixel-intensity and gradient-based methods.



Most Robust Task: Semantic Segmentation (SegFormer)

Relies on deep semantic context, ignoring low-level pixel corruption.



Best Recovery Strategy: LowLight Brightening (Gamma + CLAHE)

Recovers raw features, though detection requires clear semantics.

KEY TAKEAWAY

Pixel-level semantics survive environmental stress far better than region-level detection (YOLO) or raw edge detection.